

An informed estimate of the proton-therapy installed base over the next 10–20 years

Note prepared for the ARGOS/CRYSP proton-therapy proposal

September 2026

Abstract

We estimate the number of proton-therapy centres and treatment rooms in clinical operation worldwide in 2036 and 2046. The estimate is built from three independent inputs: the historical growth of the installed base (PTCOG lists, 2000–2026), the current construction and planning pipeline (PTCOG and IBA investor reports), and a demand ceiling derived from cancer incidence and the fraction of radiotherapy patients considered proton-eligible. Our central scenario is that the installed base roughly doubles in ten years and triples in twenty: 200–250 centres with 500–600 treatment rooms by 2036, and 300–400 centres with 700–1000 rooms by 2046, sustained by a flow of 20–30 new rooms per year. Low and high scenarios bracket this at roughly $\pm 30\%$ in 2036 and a factor of two in 2046. Demand is not the limiting factor in any scenario; system cost, reimbursement and clinical evidence are.

1 Purpose and scope

The ARGOS business case depends on two market quantities: the annual flow of new treatment rooms, which sets the new-installation market, and the stock of existing rooms, which sets the retrofit market and the recurring-service base. Independent market reports quoted in the funding deck agree that the current global proton-therapy market is of order \$1 billion per year, but they give revenue, not units, and their scopes differ. This note provides a unit-based estimate, expressed in centres and treatment rooms, with the reasoning and sources made explicit so that each assumption can be revised.

Throughout, a *centre* is a facility listed by the Particle Therapy Co-Operative Group (PTCOG) as being in clinical operation, and a *room* is a treatment room (gantry or fixed beam). The room is the natural unit for a per-room product such as ARGOS.

2 The installed base in 2026

Table 1 summarises the present state. The centre count comes from the PTCOG list of facilities in clinical operation. The room count is not published by PTCOG as a single figure; the value of about 300–310 rooms is a trade estimate that is consistent with the IBA installed-base share of 42% of rooms and its 45–47 operating sites.

Two features of the pipeline matter for the projection. First, it is geographically concentrated: China accounts for more than half of all new projects, with more than 30 centres operating or under construction [5]; Spain is building ten publicly funded systems at nine sites under a single order [8]; the United States pipeline is modest relative to its installed base of more than 45 centres. Second, about two-thirds of new projects are single-room facilities, so the average number of rooms per new centre (about 1.5) is well below the installed-base average of 2.3. Centre counts therefore grow faster than room counts.

Table 1: Proton-therapy installed base and pipeline, 2026.

Quantity	Value	Source
Centres in clinical operation	≈ 130 (128 in May 2025)	PTCOG [1]
Treatment rooms in operation	≈ 300 –310	trade estimate, IBA [6]
Rooms per centre, installed base	≈ 2.3	derived
Under construction (June 2026)	35 centres, ≈ 57 rooms, open 2026–2029	PTCOG [2]
In planning stage (January 2025)	41 centres, ≈ 50 rooms	PTCOG [3]
IBA pipeline (H1 2026)	47 operating, 11 installing, 35 in production	IBA [7]
Single-room fraction of pipeline	$\approx 65\%$	review [5]
Cumulative patients treated	$> 450,000$	review [5]

3 Historical growth

The growth of the installed base is well documented [4, 5]. Table 2 gives the series. The Lancet Oncology review describes the long-run trend as a doubling every seven years. The recent data show, however, that the exponential phase ended around 2020: the base grew by about 70 centres in 2010–2020 and by about 35 centres in 2020–2025, i.e. a roughly constant 6–8 centres per year. The United States market saturated after 2018, and growth has shifted to China, Spain, Korea, Taiwan and India.

Table 2: Proton-therapy centres in clinical operation worldwide.

Year	Centres	Growth in decade	Source
2000	10	–	[4]
2010	25	$\times 2.5$	[4]
2020	95	$\times 3.8$	[4]
2021	98		PTCOG, quoted in [5]
2023	106		[4] (March 2023)
2025	128		PTCOG [1] (May 2025)
2026	≈ 130 –135		this note

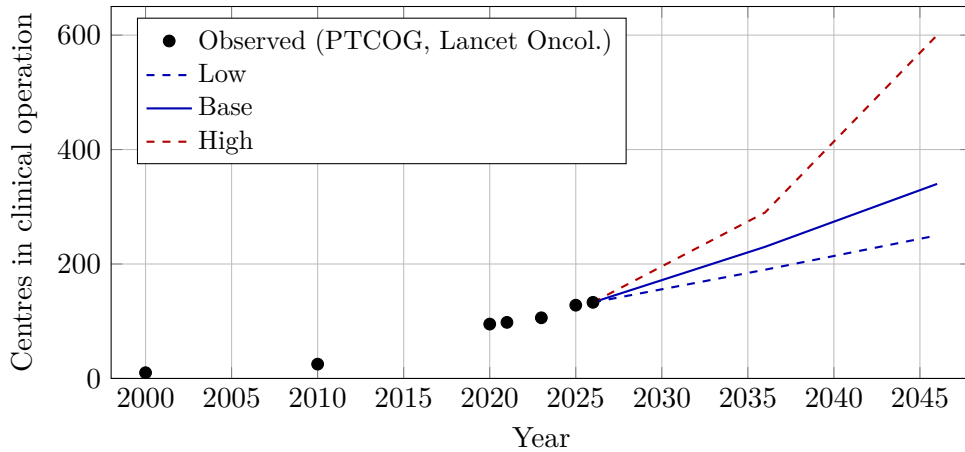


Figure 1: Observed number of proton-therapy centres and the three projection scenarios of Section 6.

4 The annual flow of new rooms

The flow of new rooms is the quantity that determines the new-installation market. IBA reports the number of rooms it sells and its share of new rooms sold, from which the global figure follows (Table 3). The IBA share accounting excludes most Chinese domestic vendors (Shanghai APACTRON, Hefei CIM, and others), which have received regulatory approval and are supplying part of the Chinese pipeline, so the true global figure is probably 20–25 rooms per year.

Table 3: New proton-therapy rooms sold, from IBA investor reports [6, 7].

Period	IBA rooms	IBA share of new rooms	Global rooms (implied)
FY2024	4	56%	≈ 7
FY2025	12	63%	≈ 19
H1 2026	5	31%	≈ 16 (half year)

Two cross-checks give the same rate. The PTCOG construction list contains about 57 rooms opening over four years, and the planning list about 50 more, which together with the Chinese domestic projects give a pipeline of 110–150 rooms over five to seven years, i.e. 20–25 rooms per year. The IBA backlog alone, 46 sites in installation or production at roughly 1.5 rooms per site, is about 70 rooms. At its long-run share of 40–60% this again implies a global pipeline of 120–170 rooms.

The value of 19 rooms per year used in the ARGOS installation-market model (file `costs/installation_market.csv`) is therefore a fair central value, but it is the 2025 figure, which was IBA’s second-best year ever. It should be quoted as a range of 15–25 rooms per year.

5 Demand ceiling

The projection should be bounded by an estimate of how many patients could plausibly be treated with protons. About half of all cancer patients require radiotherapy. The fraction of radiotherapy patients who would benefit from protons is debated: a survey of experts gave a median of 15% (range 4–92%), while the fraction agreed by 95% of respondents was 4.3% [10]. Global cancer incidence is about 20 million new cases per year, rising to 30 million by 2040. The proton-eligible population is therefore of order 0.5–2 million patients per year.

Current capacity is far below this. With about 300 rooms treating 150–250 patients per room per year (exceptional single-device centres reach 1000), the world treats roughly 40–60 thousand patients per year with protons, i.e. 0.3–0.5% of radiotherapy patients. Even the conservative 4% eligibility would require roughly ten times the present room count. Demand is not the constraint in any of the scenarios below; system cost, reimbursement and the outcome of the 35 ongoing phase III trials comparing protons with photons are [7].

A second, planning-based bound is the European Society for Radiotherapy and Oncology rule of thumb of one proton centre per about 5 million inhabitants [9]. Applied to the high-income world (1.3 billion people) this gives about 260 centres; including upper-middle-income countries (about 4 billion) it gives about 800. The installed base would reach the first figure in the 2040s under the base scenario and the second only under the high scenario.

6 Projection scenarios

The three scenarios in Table 4 start from 133 centres and 310 rooms in 2026. All assume that new centres average 1.5–1.8 rooms and that about one older room per year is retired.

Low. Linear continuation of the 2020–2026 trend: 6 net new centres and 14 net new rooms per

year. This corresponds to reimbursement pressure in the United States and Europe, with only China sustaining growth.

Base. The current pipeline is built out and the flow rises modestly: 20 rooms per year to 2030, 25 per year in 2031–2036, 30 per year thereafter. Centres grow at 8–10 per year. This is roughly 5% annual growth of the room base early in the period, slowing to 3% by the 2040s.

High. Sustained 8% annual growth of the room base, a doubling every nine years, driven by China and India scaling to national coverage and by compact single-room systems reaching regional hospitals. This requires a step change in evidence or a large drop in system cost.

Table 4: Projected proton-therapy installed base in clinical operation.

Scenario	2036		2046		New rooms
	Centres	Rooms	Centres	Rooms	per year
Low	≈ 190	≈ 450	≈ 250	≈ 590	14
Base	≈ 230	≈ 550	≈ 340	≈ 850	20–30
High	≈ 290	≈ 670	≈ 600	≈ 1400	25–60

Our informed guess is the base scenario: the installed base roughly doubles in ten years and triples in twenty. Expressed as ranges, this is 200–250 centres and 500–600 rooms in 2036, and 300–400 centres and 700–1000 rooms in 2046, with a flow of 20–30 new rooms per year throughout.

7 Consistency with revenue forecasts

The market reports quoted in the funding deck put the systems market at about \$1 billion in 2025 and forecast growth of 6–11% per year to 2030–2035. At a typical selling price of €20–30 million per room including installation, and allowing for service revenue on the installed base, a systems market of \$1–1.5 billion corresponds to 20–35 rooms per year. The unit-based base scenario is therefore consistent with the lower half of the analyst range, and the high scenario with its upper half.

8 Implications for the ARGOS market model

1. **New-installation market.** The 19 rooms per year in the current model is a reasonable central value. A sensitivity range of 15–25 rooms per year should be shown, with 20–30 as the base expectation for the 2030s.
2. **Retrofit base.** The model assumes 300 eligible existing rooms, which equals today’s entire installed base. If the retrofit window is the 2030s, the eligible base is 450–550 rooms, so the assumption is conservative provided room compatibility holds.
3. **Recurring services.** The serviced base compounds with the installed base. Under the base scenario a product that captures a constant share of new rooms sees its service base grow by about 5% per year from installations alone.
4. **Units.** Because new builds are mostly single-room, quoting the market in rooms rather than centres avoids overstating growth by roughly 30%.

9 Caveats

The room count of the installed base is a trade estimate, not a PTCOG figure. The PTCOG planning list is dated January 2025 and historically only 50–70% of planned projects are built on schedule. IBA share data exclude most Chinese domestic vendors, so the Chinese contribution is the largest single uncertainty in the flow. The demand ceiling depends on the proton-eligible fraction, which spans an order of magnitude across experts. None of these uncertainties changes the conclusion that the installed base will grow by a factor of two to three over twenty years; they mainly affect the timing.

References

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